LISP Solutions – Senior Division

1. **01-02 C1 Lisp Programming**

   CAR returns the first element of the list. The list here has 3 elements. The first element is (2 (3)). (2 (3))

2. **02-03 C1 Lisp Programming**

   OTH = ‘(10 J (Q K) A) so (CDR OTH) = ‘(J (Q K) A)
   (CDR ‘(J (Q K) A)) = ((Q K) A) so (CAR ‘((Q K) A)) = (Q K)
   (REVERSE ‘(Q K)) = (K Q) (K Q)

3. **03-04 C1 Lisp Programming**

   (SQUARE (DIV (EXP 2 3) (SUB 10 8))) = (SQUARE (DIV 8 2)) =
   (SQUARE 4) = 16 16

4. **04-05 C1 Lisp Programming**

   (CDR ‘((A B) C (D E F) (G H))) = ‘(C (D E F) (G H))
   (REVERSE ‘(C (D E F) (G H))) = ‘((G H) (D E F) C)
   (CDR ‘((G H) (D E F) C)) = ‘((D E F) C)
   (CAR ‘((D E F) C)) = (D E F) (D E F)

5. **05-06 C1 Lisp Programming**

   (CDR ’(3 (4 1 (3 2)) 1 (2 (1 3) 4))) =’((4 1 (3 2)) 1 (2 (1 3) 4)) ‘((3 2))
   (CAR ’((4 1 (3 2)) 1 (2 (1 3) 4)) = ’(4 1 (3 2))
   (CDR ’(4 1(3 2)) = ’(1 (3 2))
   (CDR ’(1 (3 2)) = ’((3 2))

6. **06-07 C1 Lisp Programming**

   (CDR’ (2 (3 1) (4 5 (2 3)) (6 1 (8 2)))) = ((3 1) (4 5 (2 3)) (6 1 (8 2))) (5 (2 3))
   (CDR ’((3 1) (4 5 (2 3)) (6 1 (8 2)))) = ((4 5 (2 3)) (6 1 (8 2)))
   (CAR ’((4 5 (2 3)) (6 1 (8 2))))= (4 5 (2 3))
   (CDR ’(4 5 (2 3))) = (5 (2 3))

7. **07-08 C1 Lisp Programming**

   (CDR (CAR (CDR (REVERSE (CDR ‘((2 3) (1 2) ((2 3) 1) 2)))))) (1)
   = (CDR (CAR (CDR (REVERSE ‘((1 2) ((2 3) 1) 2)))))
   = (CDR (CAR (CDR ‘(2 ((2 3) 1) (1 2)))))
   = (CDR (CAR ‘(((2 3) 1) (1 2))))
   = (CDR ‘((2 3) 1)) = (1)

8. **08-09 C1 Lisp Programming**

   (CAR (REVERSE (CDR ‘(3 2 (4 (1 2)) 7 6 (2 1))))) (2 1)
   = (CAR (REVERSE ‘(2 (4 (1 2)) 7 6 (2 1))))
   = (CAR ‘((2 1) 6 7 (4 (1 2)) 2))
   = (2 1)

9. **09-10 C1 Lisp Programming**

   (CDR (CAR (REVERSE (CDR’((8 2) 3 ((8 1) 7) 2 (1 (2 3))))))) ((2 3))
   = (CDR (CAR (REVERSE’(3 ((8 1) 7) 2 (1 (2 3))))))
   = (CDR (CAR ’((1 (2 3)) 2 ((8 1) 7) 3)))
   = (CDR ’(1 (2 3)))
   = ((2 3))

11 11-12 C2 Lisp Programming

(EXP 2 (DIV (MULT 8 6) (SQUARE 4))) 8
= (EXP 2 (DIV 48 16))
= (EXP 2 3)
= 2↑3 = 8

12. 11-12 C4 Lisp Programming

(CAR (CDR (CAR (CDR '((1 2) (3 (4 5)) 4 (3 2) 1))))) (4 5)
= (CAR (CDR (CAR '((3 (4 5)) 4 (3 2) 1))))
= (CAR (CDR '(3 (4 5))))
= (CAR '((4 5)))
= (4 5)

13. 12-13 C2 Lisp Programming

(SUB (ADD 2 (MULT 5 2) 3) (DIV 9 (EXP 3 1))) 12
= (SUB (ADD 2 (MULT 5 2) 3) (DIV 9 3))
= (SUB (ADD 2 10 3) 3)
= (SUB 15 3)
= 12

14. 12-13 C4 Lisp Programming

(CAR (CAR (CDR (REVERSE (CDR '((2 (1 2)) (4 1 3) ((2 1) 3) 4)))))) (2 1)
= (CAR (CAR (CDR (REVERSE '((4 1 3) ((2 1)3) 4)))))
= (CAR (CAR (CDR '(4 ((2 1)3) (4 1 3)))))
= (CAR (CAR '(((2 1) 3) (4 1 3))))
= (CAR '((2 1) 3)) = (2 1)

15. 13-14 C2 Lisp Programming

(MULT (ADD (SUB 10 2) 3) (EXP 2 4)) 176
= (MULT (ADD 8 3) 16)
= (MULT 11 16) = 176

16. 13-14 C4 Lisp Programming

(CAR (CAR (REVERSE '((A B) C ((D E) F))))) (D E)
= (CAR (CAR '(((D E) F) C (A B))))
= (CAR '((D E) F))
= (D E)

17. 14-15 C2 Lisp Programming
(ADD (MULT 3 (EXP 2 3)) (MULT (EXP 2 2) (SUB (MULT 5 3) 6)) 124
(EXP 2 (SUB (9 3))))
= (ADD (MULT 3 8) (MULT 4 (SUB 15 6)) (EXP 2 6))
= (ADD 24 (MULT 4 9) 64)
= (ADD 24 36 64)
= 124

18. 14-15 C4 Lisp Programming

(REVERSE (CAR (CDR (CDR (CAR '((a b (c d)) (e (f g)) (h (j k)))))))) '(d c)
= (REVERSE (CAR (CDR (CDR '(a b (c d))))))
= (REVERSE (CAR (CDR '(b (c d)))))
= (REVERSE (CAR '((c d))))
= (REVERSE '(c d))
= '(d c)

19. 15-16 C2 Lisp Programming

(ADD (MULT 2 5) (EXP 3 2) (SUB 9 (ADD 1 7)) 39
(DIV 15 3) (ADD 4 (SUB 8 4) 6))
= (ADD 10 9 (SUB 9 8) 5 (ADD 4 4 6)) = (ADD 10 9 1 5 14) = 39

20. 15-16 C4 Lisp Programming

(CDR (CAR (CDR (REVERSE (CAR (CDR '(1 (2 ((3 4) 2) 3) 2 (4 (1 2)))))))) (2)
= (CDR (CAR (CDR (REVERSE (CAR '((2 ((3 4) 2) 3) 2 (4 (1 2))))))))
= (CDR (CAR (CDR (REVERSE '((2 ((3 4) 2) 3))))))
= (CDR (CAR (CDR '((3 ((3 4) 2) 2)))))
= (CDR (CAR '((((3 4) 2) 2)))) = (CDR '((3 4) 2)) = (2)